Default Mask
(Network Mask)

Binary Notation: Base 2

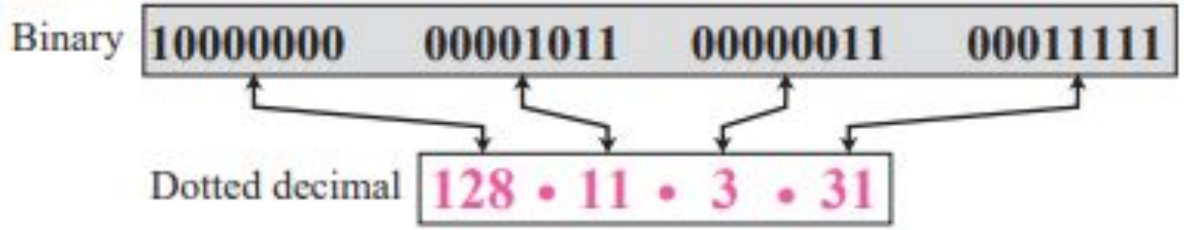
In **binary notation,** an IPv4 address is displayed as 32 bits. To make the address more readable, one or more spaces is usually inserted between each octet (8 bits). Each octet is often referred to as a byte. So it is common to hear an IPv4 address referred to as a 32-bit address, a 4-octet address, or a 4-byte address. The following is an example of an IPv4 address in binary notation:

```
01110101 10010101 00011101 11101010
```

Dotted-Decimal Notation: Base 256

To make the IPv4 address more compact and easier to read, an IPv4 address is usually written in decimal form with a decimal point (dot) separating the bytes. This format is referred to as **dotted-decimal notation.** Figure 5.1 shows an IPv4 address in dotted-decimal notation. Note that because each byte (octet) is only 8 bits, each number in the dotted-decimal notation is between 0 and 255.

Figure 5.1 *Dotted-decimal notation*



Example 5.1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

- **a.** `10000001 00001011 00001011 11101111`
- **b.** `11000001 10000011 00011011 11111111`
- **c.** `11100111 11011011 10001011 01101111`
- **d.** `11111001 10011011 11111011 00001111`

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation:

- **a.** 129.11.11.239
- **b.** 193.131.27.255
- **c.** 231.219.139.111
- **d.** 249.155.251.15

Figure 5.6 *Finding the class of an address*

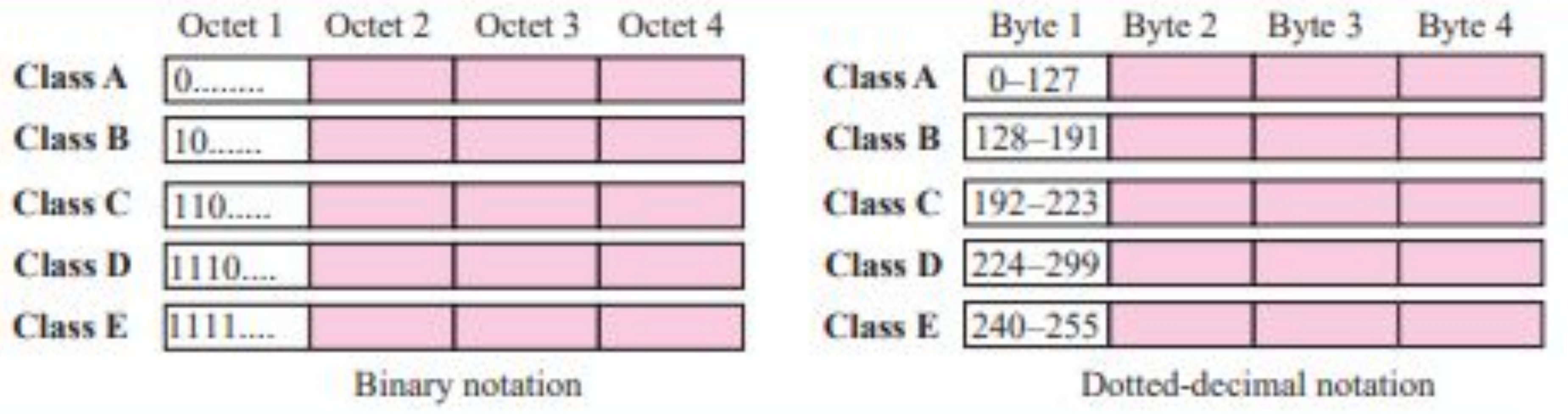
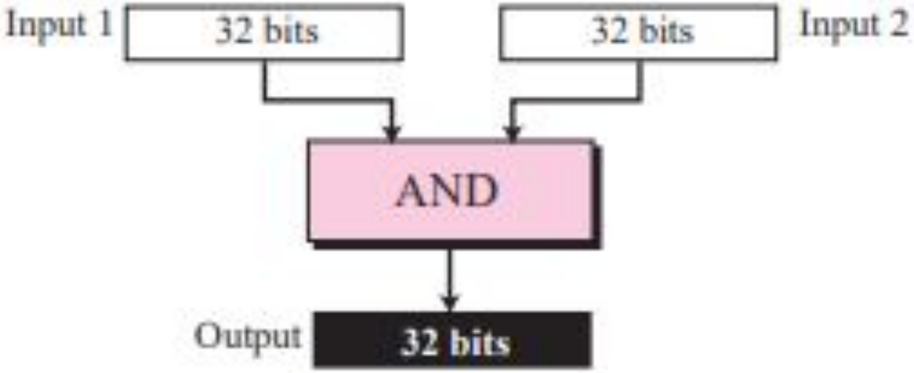


Figure 5.3 *Bitwise AND operation*

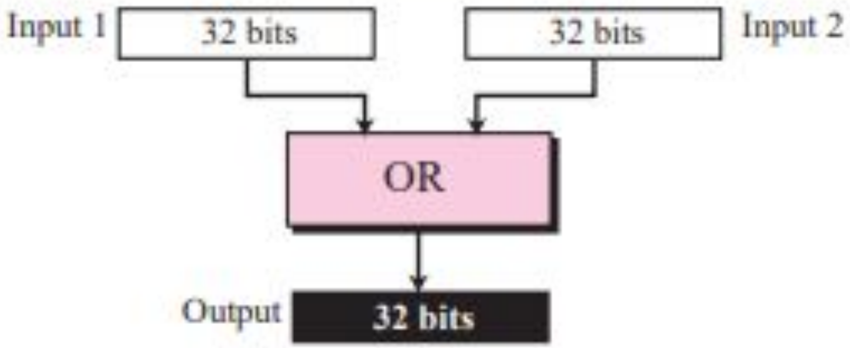


AND

Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

Operation for each bit

Figure 5.4 *Bitwise OR operation*



OR

Input 1	Input 2	Output
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**1**

Operation for each bit

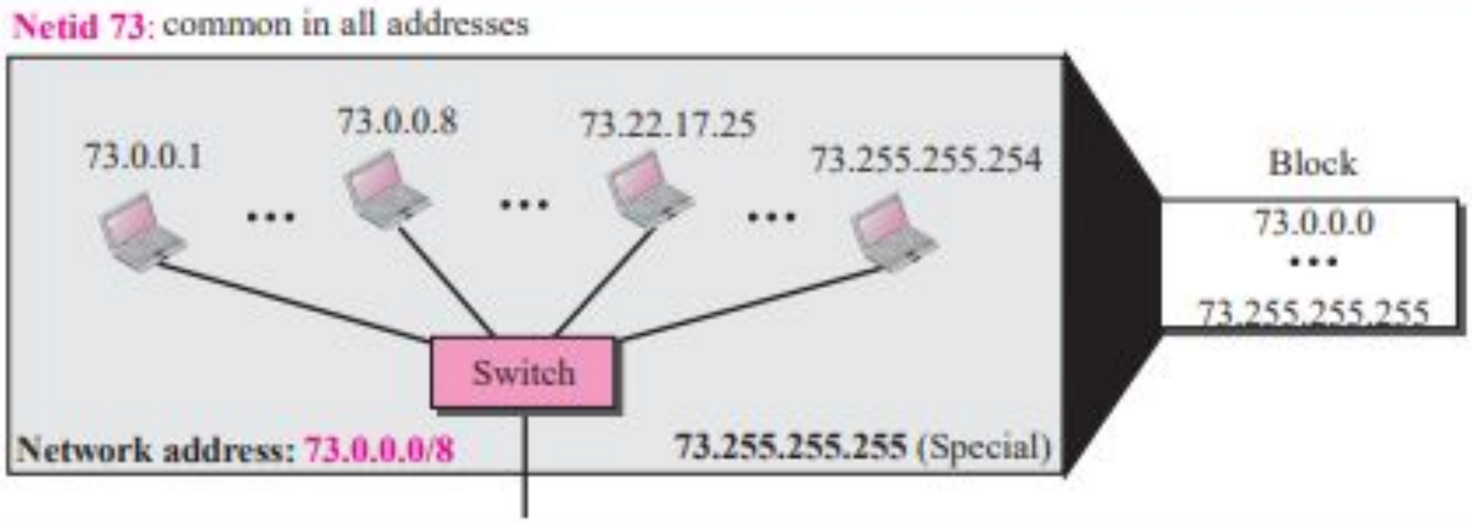
Example 5.13

An address in a block is given as 73.22.17.25. Find the number of addresses in the block, the first address, and the last address.

Solution

Since 73 is between 0 and 127, the class of the address is A. The value of n for class A is 8. Figure 5.16 shows a possible configuration of the network that uses this block. Note that we show the value of n in the network address after a slash. Although this was not a common practice in classful addressing, it helps to make it a habit in classless addressing in the next section.

Figure 5.16 *Solution to Example 5.13*



Example 5.14

An address in a block is given as 180.8.17.9. Find the number of addresses in the block, the first address, and the last address.

Solution

Since 180 is between 128 and 191, the class of the address is B. The value of n for class B is 16. Figure 5.17 shows a possible configuration of the network that uses this block.

Figure 5.17 *Solution to Example 5.14*

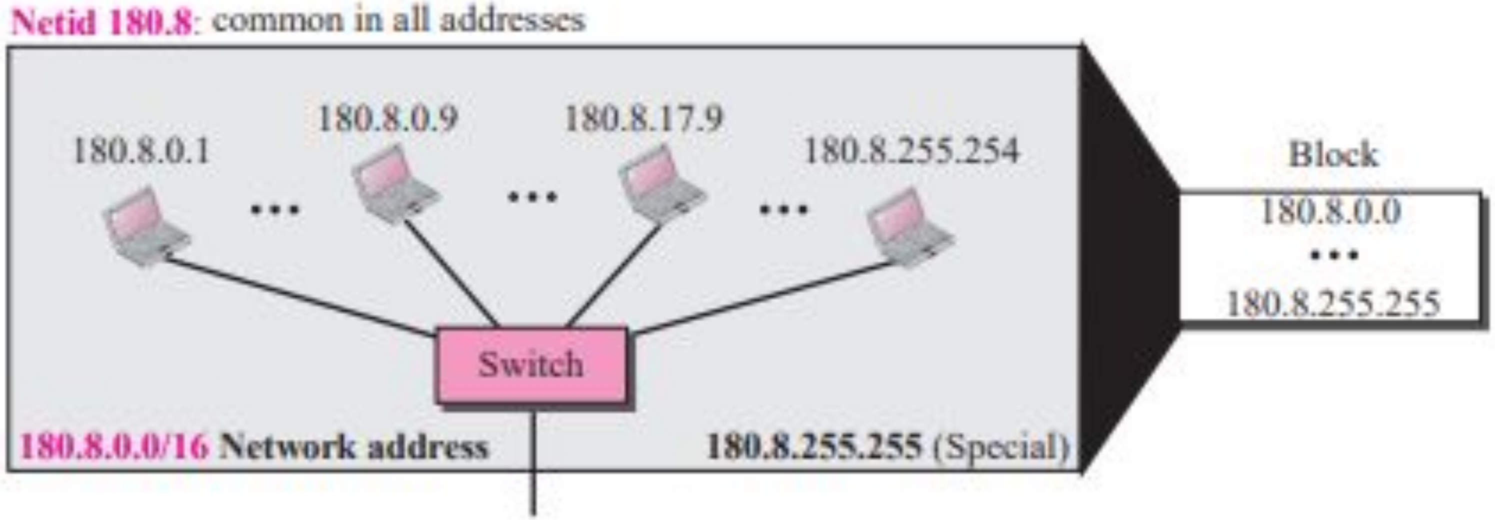
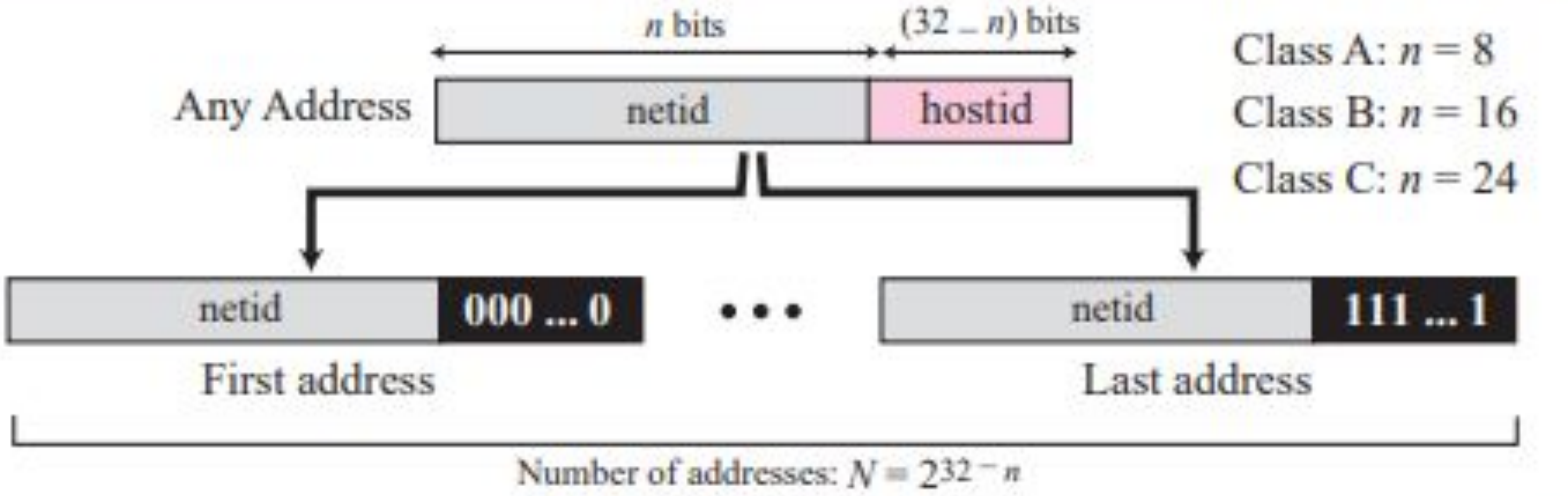


Figure 5.15 *Information extraction in classful addressing*



Default Subnet Mask

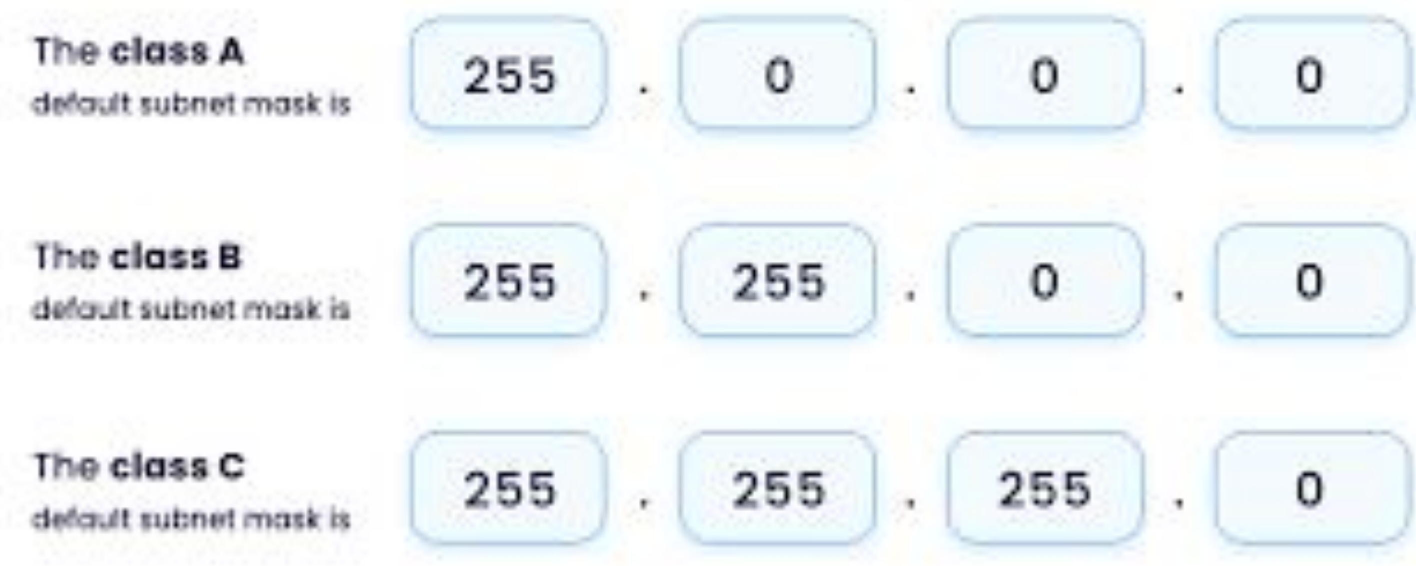
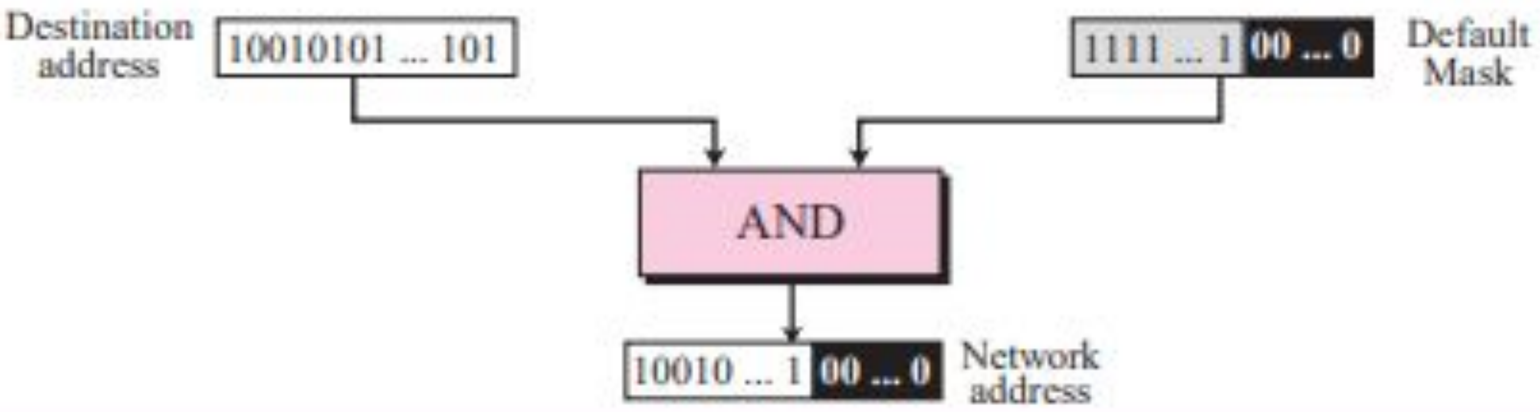


Figure 5.22 *Finding a network address using the default mask*



Example 5.16

A router receives a packet with the destination address 201.24.67.32. Show how the router finds the network address of the packet.

Solution

We assume that the router first finds the class of the address and then uses the corresponding default mask on the destination address, but we need to know that a router uses another strategy as we will discuss in the next chapter. Since the class of the address is B, we assume that the router applies the default mask for class B, 255.255.0.0 to find the network address.

Destination address	→	201	.	24	.	67	.	32
Default mask	→	255	.	255	.	0	.	0
Network address	→	201	.	24	.	0	.	0

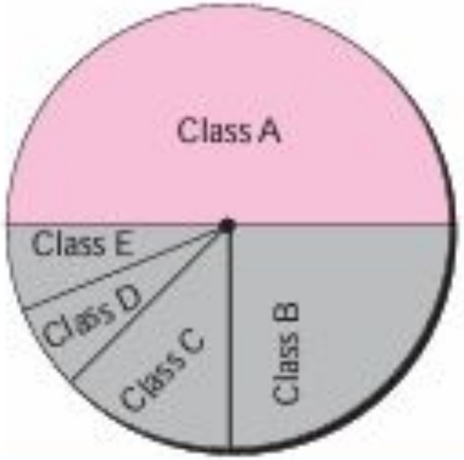
We have used the first short cut as described in the previous section. The network address is 201.24.0.0 as expected.

Subnet Mask

Classes

In classful addressing, the IP address space is divided into five **classes: A, B, C, D,** and **E.** Each class occupies some part of the whole address space. Figure 5.5 shows the class occupation of the address space.

Figure 5.5 *Occupation of the address space*



Class A: 2^{31} = 2,147,483,648 addresses, 50%

Class B: 2^{30} = 1,073,741,824 addresses, 25%

Class C: 2^{29} = 536,870,912 addresses, 12.5%

Class D: 2^{28} = 268,435,456 addresses, 6.25%

Class E: 2^{28} = 268,435,456 addresses, 6.25%

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

Example 5.18

Figure 5.23 shows a network using class B addresses before subnetting. We have just one network with almost 2^{16} hosts. The whole network is connected, through one single connection, to one of the routers in the Internet. Note that we have shown /16 to show the length of the netid (class B).

Figure 5.23 *Example 5.18*

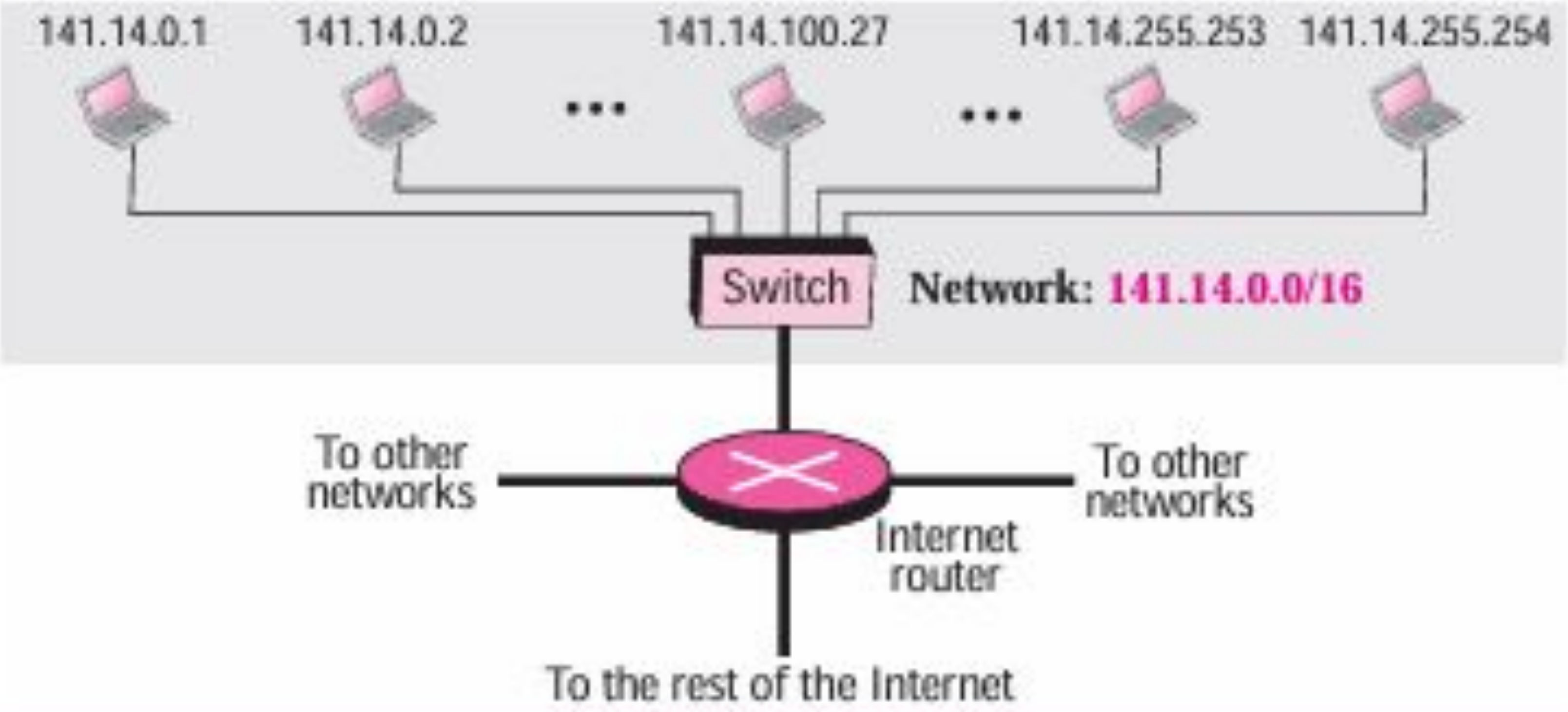


Figure 5.24 *Example 5.19*

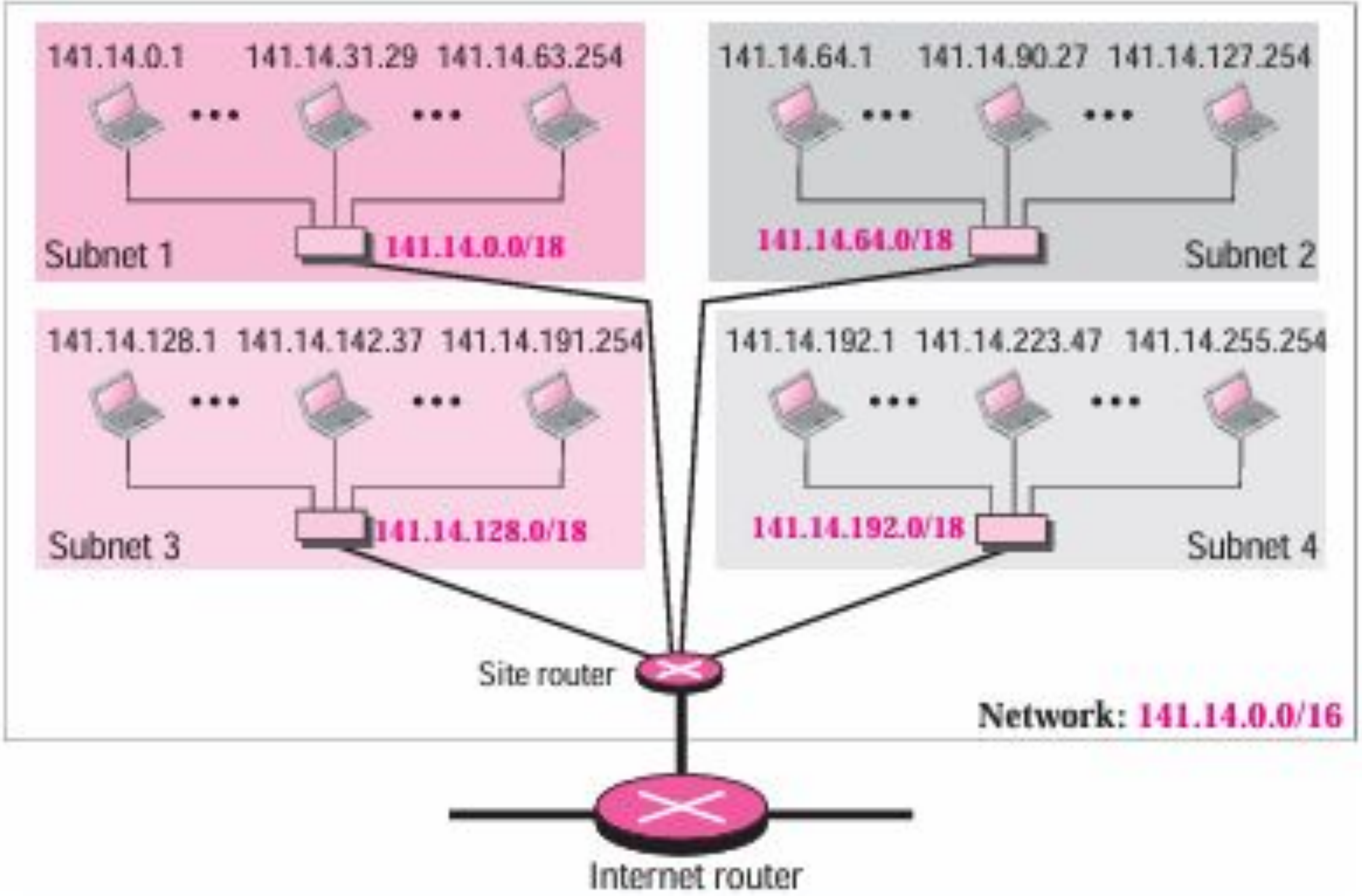
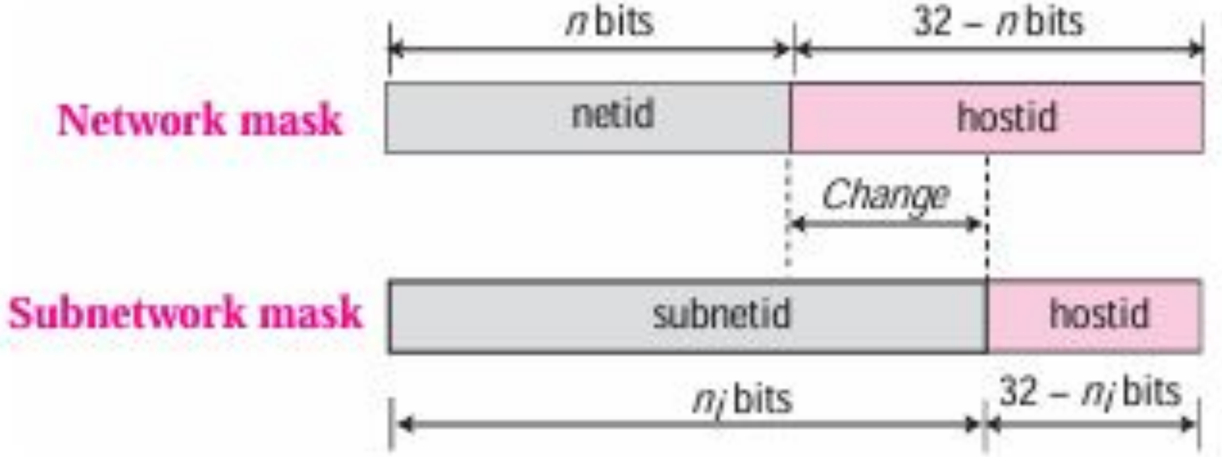


Figure 5.25 *Network mask and subnetwork mask*



Subnetting increases the length of the netid and decreases the length of hostid. When we divide a network to s number of subnetworks, each of equal numbers of hosts, we can calculate the subnetid for each subnetwork as

$$n_{sub} = n + \log_2 s$$

in which n is the length of netid, n_{sub} is the length of each subnetid, and s is the number of subnets which must be a power of 2.

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in which n is the length of netid, n_{sub} is the length of each subnetid, and s is the number of subnets which must be a power of 2.

Example 5.20

In Example 5.19, we divided a class B network into four subnetworks. The value of $n = 16$ and the value of $n_1 = n_2 = n_3 = n_4 = 16 + \log_2 4 = 18$. This means that the subnet mask has eighteen 1s and fourteen 0s. In other words, the subnet mask is 255.255.192.0 which is different from the network mask for class B (255.255.0.0).

Example 5.21

In Example 5.19, we show that a network is divided into four subnets. Since one of the addresses in subnet 2 is 141.14.120.77, we can find the subnet address as:

Address	→	141	.	14	.	120	.	77
Mask	→	255	.	255	.	192	.	0
Subnet Address	→	141	.	14	.	64	.	0

If you expect to have 100 remote sites with 300 PCs each, with the network address 192.168.0.0, what subnet mask should you use?

255.255.0.0 = 11111111.11111111.00000000.00000000

255.255.254.0 = 11111111.11111111.11111110.00000000

The new mask uses seven subnet bits, which means *s* equals *7*, so you need to calculate 2^7.

$2^7 = 128$

Calculating subnets and hosts

This table shows the number of subnets and hosts for each of eight mask bits in the third octet of an IPv4 address.

MASK	SUBNETS (2^s)	HOSTS (2^h-2)
255.255.0.0	1	65,534
255.255.128.0	2	32,766
255.255.192.0	4	16,382
255.255.224.0	8	8,190
255.255.240.0	16	4,094
255.255.248.0	32	2,046
255.255.252.0	64	1,022
255.255.254.0	128	510
255.255.255.0	256	254

Example 1: IP address: 10.0.0.55 and Subnet Mask:255.255.255.0

Bitwise AND:

00001010.00000000.00000000.00110111

11111111.11111111.11111111.00000000

Subnet ID: 00001010.00000000.00000000.00000000

Therefore: Subnet ID: 10.0.0.0 and Host ID:55

Example 2: IP address: 172.16.14.101 and Subnet Mask:255.255.248.0

Bitwise AND:

10101100.00010000.00001110.01100101

11111111.11111111.11111000.00000000

Subnet ID: 10101100.00010000.00001000.00000000

Therefore: Subnet ID: 172.16.8.0 and Host ID:14.101